News Articles Classification

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***Abstract*—This report investigates the multi-class classification of news articles data. The dataset is processed through a strict cleaning phase, together with hybrid TF-IDF strategy combining word and character-level n-grams. By comparing Multinomial Naive Bayes and Linear Support Vector Classifiers, the study highlights how the latter better exploits the high-dimensional feature space provided by the character-word synergy. The results demonstrate that the Linear SVC, when paired with balanced class weights and carefully selected hyperparameters, consistently outperforms the Naive Bayes configuration, achieving a satisfactory classification performance.**

I. Problem overview

The dataset provided for this study consists of a large-scale collection of web-scraped news articles, a characteristic that is evident from the high presence of URL fragments and HTML elements related to embedded images and media content. Each sample is characterized by a heterogeneous set of features, including the article *title*, the complete *article* body, the publishing *source*, and auxiliary numerical attributes representing the source *page rank* and the publication *timestamp*.

The corpus is structured into two primary subsets:

- **Development set**: comprising 79,997 observations, each associated with one of seven distinct thematic categories, and utilized for model training, feature engineering, and cross-validation;
- **Evaluation set**: consisting of 20,000 unlabeled samples, intended for final performance assessment and submission.

The core objective is to build a classification system capable of mapping the textual content of an article to its corresponding label. As defined in the task specifications, the taxonomy includes seven classes: International News (0), Business (1), Technology (2), Entertainment (3), Sports (4), General News (5), and Health (6).

The main challenges of this task reside in the inherent noise of web-scraped data—such as HTML boilerplate, potentially duplicated content across sources, and URL fragments—and in the requirement to optimize the Macro-F1 score, which enforces balanced predictive performance across both majority and minority classes.

II. Proposed approach

A. Preprocessing

An initial feature-level analysis revealed that the textual attributes *article* and *title*, together with the categorical feature *source*, had the strongest impact on the model's predictive performance. Consequently, the preprocessing pipeline was primarily designed to clean, normalize, and strengthen these features, while mitigating sources of noise typical of web-scraped content. The other features were also retained, as they slightly improved the model's performance without significantly increasing dimensionality.

1) Cleaning: A preliminary inspection of the development set revealed only two records containing missing values in the *title* or *article* fields, which were therefore removed. A more in-depth analysis, however, uncovered a non-negligible level of data redundancy: approximately 7% of the samples corresponded to duplicated articles, likely originating from news syndication across multiple sources. To prevent potential *data leakage* during cross-validation—where identical samples could appear in both training and validation folds, duplicate entries were also removed, retaining only the first occurrence of each unique article.

TABLE I
Class Distribution in the Development Set After Deduplication

Category	Count	Percentage (%)
International News	22,723	30.54
Business	9,405	12.64
Technology	10,769	14.48
Entertainment	9,346	12.56
Sports	8,425	11.32
General News	10,832	14.56
Health	2,893	3.89

After this initial cleaning step, the class distribution of the development set remained noticeably imbalanced. As summarized in Table I, *International News* constitutes the majority class, while *Health* is strongly under-represented.

The textual content of articles and titles was cleaned using regular expressions: URLs were removed, while *alt* attributes were extracted to retain meaningful textual information. Titles and article bodies were then concatenated, converted to lowercase, and excess whitespace was normalized. This process also contributed to reducing the vocabulary size and mitigating noise introduced by URLs and special symbols. Additional attempts to preserve informative signals were explored, such as replacing monetary symbols with the token "moneyamount"; however, this strategy resulted in only marginal performance improvements, suggesting that lexical context plays a more central role in the classification task.

Finally, very short articles were removed, as they contained little information, contributed noise, and were unlikely to

provide meaningful input for the model.

2) Feature extraction:

- **Textual features (TF-IDF).** The content of the concatenated *title* and *article* fields was first stripped of stopwords and then lemmatized, in order to reduce inflectional variability while preserving semantic meaning. The resulting text was vectorized using a word-level TF-IDF representation, yielding a vocabulary of 60,866 terms. Additionally, a character-level TF-IDF representation was incorporated as an auxiliary feature, which proved particularly beneficial for the Linear SVC model in capturing subword patterns and fine-grained textual information.
- **Source encoding.** The *source* attribute was treated as a categorical feature and encoded accordingly, allowing the model to exploit source-specific publishing patterns.
- **Article length.** The distribution of article lengths across categories is shown in Fig. 1. Due to the heavy-tailed nature of the data, a logarithmic scale is adopted. While the median article length is broadly comparable across classes, *Technology* articles display higher variability and a larger number of extreme outliers. This indicates that article length alone is not a strong discriminative feature, but it may provide weak complementary information, particularly for unusually long documents. For this reason, article length was retained as an additional numerical feature in the preprocessing pipeline. Empirical evaluation, however, showed that its inclusion resulted only in marginal performance variations and did not significantly improve the overall Macro-F1 score.

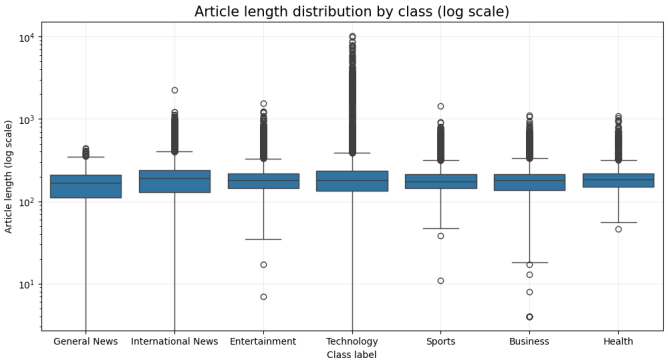

Fig. 1. Distribution of article length by category (log scale).

- **Other information.** Temporal metadata and page rank were included as additional numerical features. The publication year was extracted from the article timestamp and one-hot encoded, with a special category for non-parsable values (about 25,000 articles). Although neither feature produced substantial improvements, both provided minor performance gains and were retained in the preprocessing pipeline.

B. Model selection

Based on previous studies, I selected two widely used text classification models: Linear Support Vector Classifier (Linear SVC) and Multinomial Naive Bayes (Multinomial NB). Both algorithms have been shown to perform effectively on multi-class and multi-label text datasets [1], [2], motivating their adoption in this study.

- **Multinomial Naive Bayes.** Multinomial Naive Bayes represents a common baseline for text classification tasks, as it effectively exploits word occurrence statistics in sparse representations such as TF-IDF vectors. However, its strong conditional independence assumptions and its sensitivity to extremely high-dimensional feature spaces limit its ability to fully leverage complex feature combinations.
- **Linear Support Vector Classifier.** The Linear SVC proved more suitable for the adopted feature representation. Owing to its margin-based optimization and robustness in high-dimensional spaces, it is able to better exploit the rich textual representations produced by the combined word-level and character-level TF-IDF features, as well as the additional structured attributes.

C. Hyperparameters tuning

To identify the optimal configuration for both feature extraction and classification, an extensive search was performed. The parameters and their respective search spaces are summarized in Table II.

TABLE II
HYPERPARAMETER SEARCH SPACE FOR FEATURE EXTRACTION AND MODEL SELECTION

Phase	Model	Parameter	Value Range / Set
Preprocessing	TF-IDF (Word)	ngram_range	(1, 2), (1, 3)
		max_df	$0.01 \rightarrow 0.09$
		min_df	$1 \rightarrow 10$
		use_idf	[True, False]
		sublinear_tf	[True, False]
		binary	[True, False]
	TF-IDF (Char)	**Usage**	**[Included, Excluded]**
		ngram_range	(3, 4), (3, 5), (4, 6)
		max_df	$0.01 \rightarrow 0.09$
		min_df	$1 \rightarrow 10$
		use_idf	[True, False]
		sublinear_tf	[True, False]
Model Selection	Linear SVC	C	$0.01 \rightarrow 10$
		class_weight	[None, 'balanced']
	Multinomial NB	alpha	$0.01 \rightarrow 10$
		fit_prior	[True, False]

During the tuning phase, a clear divergence emerged in the effect of character-level n-grams on the two candidate models. While the Linear SVC benefited from the structural information captured by character sequences (3–5 grams), the performance of Multinomial Naive Bayes significantly degraded when these features were included. This behavior can be attributed to the strong correlation among overlapping character n-grams, which violates the conditional independence assumption underlying the Naive Bayes classifier. As a result, the search space for Multinomial NB was restricted to word-level TF-IDF features only.

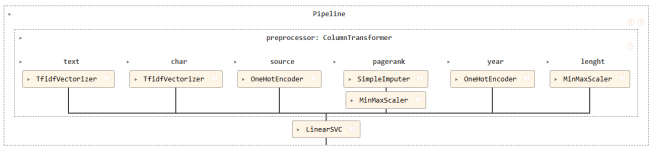

Fig. 2. Full architecture of the proposed machine learning pipeline.

III. Results

The Linear Support Vector Classifier emerged as the best-performing model in this study. The optimal configuration was automatically identified through an extensive grid search. Interestingly, the resulting parameters align with our expectations, as they capture both rich textual information and robust regularization. The details are summarized below:

- **Word-level TF-IDF** (parameters selected by grid search)
  - *ngram_range = (1, 3)*: includes unigrams, bigrams, and trigrams, allowing the model to capture both simple and slightly more complex word patterns.
  - *sublinear_tf = True*: reduces the effect of very frequent terms, which is expected to improve generalization.
  - *max_features = 200,000*: the large vocabulary captures rich semantic content; balanced by min_df and max_df to avoid dimentionality issues.
  - *min_df = 5*: filters out rare words that may be noisy or uninformative.
  - *max_df = 0.15*: excludes overly common words that appear in almost all documents, increasing discriminative power.
  - *use_idf = True*: emphasizes terms across the text, giving more weight to the initial words, which are likely more informative; the title was placed before the article body to allow this parameter to better capture key terms.
- **Character-level TF-IDF** (parameters selected by grid search)
  - *ngram_range = (3, 5)*: captures subword patterns, which helps with noisy text, spelling variations, and small morphological differences.
  - *sublinear_tf = True*: reduces dominance of frequent character sequences, improving balance between common and rare patterns.
  - *max_features = 120,000*: sufficiently large to capture informative subword patterns while keeping dimensionality manageable.
  - *min_df = 5, max_df = 0.06*: balances inclusion of relevant subword patterns and filters out rare or overly frequent sequences.
  - *use_idf = True*: highlights distinctive subword patterns across documents.
- **Linear SVC classifier** (parameters selected by grid search)
  - *C = 0.075*: moderate regularization; prevents overfitting while allowing the model to learn complex textual patterns.
  - *class_weight = 'balanced'*: compensates for the class imbalance in the dataset, ensuring minority classes are considered.

TABLE III
MACRO-F1 SCORES PER FOLD AND OVERALL EVALUATION. THE LAST ROW IS RESERVED FOR THE SUBMISSION MACRO-F1.

Fold	Macro-F1 score
Fold 1	0.6974
Fold 2	0.7003
Fold 3	0.6986
Fold 4	0.6951
Fold 5	0.7012
Mean	0.6985
Submission	0.715

As shown in Table III, the mean Macro-F1 score across the 5 cross-validation folds is 0.6985, while the final submission achieved a higher score of 0.715. This improvement may suggest that the evaluation dataset was somewhat easier, or that it contained partial duplicates or overlaps with the development set, which facilitated the model's predictions.

Fig. 3. Confusion matrix with precision and recall

The confusion matrix, illustrated in Fig. 3, provides a granular view of the model's performance and highlights the semantic challenges inherent in the dataset. The strong diagonal trend confirms the Linear SVC's high discriminative power, particularly for the Sports, Technology and health categories. These classes exhibit specialized vocabularies that are easily captured by the feauture space

However, a critical inspection of the off-diagonal elements reveals systematic misclassifications that reflect genuine lexical overlaps:

Entertainment and General News Noise: A significant source of confusion is observed within the Entertainment and General News categories. Unlike Sports, which relies on a highly specialized lexicon, these two classes exhibit high intra-class variance and a vast range of overlapping topics. Entertainment news often intersects with celebrity culture, legal issues, or business-related box-office reports, while General News acts as a "catch-all" for diverse events. This thematic breadth dilutes the semantic signal, causing these samples to be frequently misclassified across multiple other categories.

General News vs. International News: The model frequently confuses these categories. Given that International News represents nearly one-third of the dataset, its broad vocabulary often encompasses the topics found in General News. This lack of clear thematic boundaries suggests that "General News" often serves as a noisy subset of the more statistically dominant international reports.

Minority Class Dispersion (Health): Despite the use of *balanced* class weights, the Health category exhibits the highest rate of dispersion. This indicates that while the SVC is penalized more heavily for missing minority samples, the specific semantic signal of Health is frequently overshadowed by the more pervasive lexical patterns of the larger categories, particularly those related to global policy or general reporting.

Ultimately, the matrix confirms that the Macro-F1 score of approximately 0.6936 is not due to random error, but to these specific, linguistically complex overlaps that represent the "upper bound" of performance for a non-contextual, purely text-based classifier.

IV. Discussion

A. Dimensionality

It is noteworthy that the combination of word-level and character-level TF-IDF representations induces a very high-dimensional feature space. Although larger vocabularies can increase expressiveness, they also introduce sparsity and computational overhead, potentially harming models sensitive to dimensionality. For this reason, an explicit analysis of the vocabulary size was conducted by varying the *max_features* parameter. The analysis was structured as follows: (i) for Linear SVC, both word-level and character-level TF-IDF representations were jointly considered for simplicity, enforcing the same maximum vocabulary size n for both vectorizers ($n_{\text{word}} = n_{\text{char}} = n$); (ii) for Multinomial Naive Bayes, only word-level TF-IDF features were retained to preserve model stability.

As illustrated in Fig. 4, the performance curves exhibit convergence as the feature space size increases. The shaded regions, representing the minimum and maximum scores achieved across the 5-fold cross-validation, indicate that the Linear SVC gains stability increasing the dimensionality. This stability confirms that the model effectively handles the increasing sparsity without suffering from significant overfitting. In contrast, the Multinomial Naive Bayes exhibits high variability in performance scores, highlighting its sensitivity to the very high-dimensional and sparse feature space.

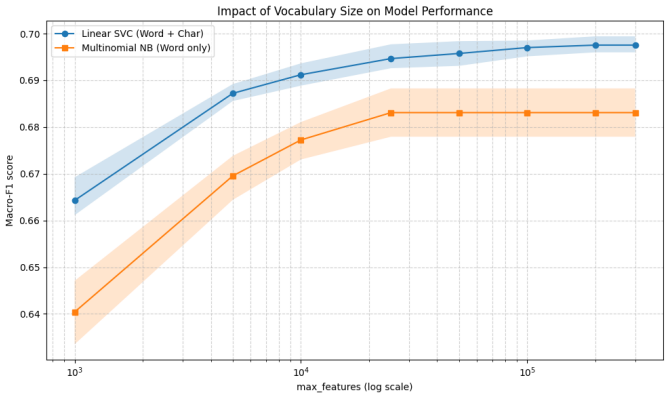

Fig. 4. Impact of Vocabulary Size on Model Performance (log scale).

B. Enforcing text signals

A possible avenue for improvement could be the development of more intensive text mining techniques to extract stronger signals, such as sports scores (e.g., 1-1), monetary amounts, or the application of feature selection methods such as chi-squared tests. Unfortunately, I was not able to fully implement this type of signal extraction, but I believe it could significantly improve the classification performance for the more challenging categories.

References

[1] S. A. Hasan, M. G. Hussain, J. Protim, M. M. Rahman, N. Fahim, and M. Z. Chowdhury, "Classification of multi-labeled text articles with reuters dataset using svm," in *2022 International Conference on Science and Technology (ICOSTECH)*, (Batam City, Indonesia), IEEE, 2022.

[2] A. McCallum and K. Nigam, "A comparison of event models for naive bayes text classification," in *AAAI Technical Report WS-98-05*, 1998. Compilation copyright © 1998, AAAI.